

Human Preferences and AI Alignment

Literature Review

Preferences in AI

Aitana Carracedo Cidoncha **Advisor:** Paul Bachmann

Supervisor: Prof. Stefan Woltran

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Abstract

AI alignment with human preferences has become central to the development of large language models. The dominant approach represents these preferences as a mathematical function, namely a utility function or reward function, and trains the model to maximise it. This literature review examines the relationship between human preferences and current alignment methods, organised around two guiding questions: to what extent current methods capture or respect real human preferences, and whether better methods can resolve the remaining problems or whether there are fundamental limits in the capacity of machine learning to represent diverse, changing, and conflicting preferences. The review finds that the literature identifies conceptual, technical, and empirical limits of this approach: the notion of preference is too thin to represent what people value, a single reward function cannot capture a diverse society, and there is evidence that current alignment methods measurably reduce pluralism. In response, overlapping alternatives have been proposed, including pluralistic alignment, explicit aggregation through social choice, and approaches that shift the target of alignment toward norms and roles. None of them resolves the problem in full, because the core of the question is normative rather than technical. The review concludes that what an AI should align with cannot be answered by more data or better algorithms alone. Open gaps remain in representing heterogeneity, handling preferences that change across time and context, and defining the limits of user agency.

1 Introduction

We have become accustomed to using artificial intelligence systems on a daily basis. Most of the time we do so without asking whether their responses reflect our values, or what we even mean by “human values” when we speak of aligning an AI. For years this question remained a theoretical concern discussed within a small research community. With the growing use of large language models, it has become practically relevant. The field of AI alignment, which studies how to build systems that behave in accordance with what humans want, has emerged as a response.

As Gabriel [18] observes, the expression “aligning AI with human values” can refer to different targets: a user’s literal instructions, their stated intentions, their revealed preferences, the preferences they would hold if better informed, their objective interests, or the moral values that the system should respect [18, Section 3]. These targets do not always coincide, and any concrete method has to take a position on which of them counts, even when it does not say so explicitly.

The dominant approach in current practice resolves this ambiguity in a specific way: it represents human preferences as a mathematical function and trains the model to maximise it. Zhi-Xuan et al. [47] call this the *preferentist* approach to AI alignment, and identify it as the framework behind the methods most commonly used today to train large language models, such as Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimisation (DPO).

This review examines that paradigm. The work is organised around two guiding questions:

- To what extent do current methods capture or respect real human preferences?
- Can better methods resolve the remaining problems, or are there fundamental limits in the capacity of machine learning to represent diverse, changing, and conflicting preferences?

The papers reviewed here were selected through a targeted literature review. The starting point was three papers assigned as core readings for this project: Zhi-Xuan et al. [47], Gözl, Haghtalab, and Yang [19], and Sorensen et al. [44]. From these, I identified further work through citation chasing, following references in these papers and in related surveys, especially Casper et al. [9] and Conitzer et al. [12]. The bibliographies of key papers, in particular those of Siththaranjan, Laidlaw, and Hadfield-Menell [42], Gözl, Haghtalab, and Yang [19], and Kirk et al. [27], were a major source of additional references. I also searched Google Scholar and arXiv for specific authors, methods, or topics when a gap in the review needed to be addressed. Philosophical and decision-theoretic sources, such as work on bounded rationality, prospect theory, and money-pump arguments, were identified through the references in the AI alignment literature itself. I included papers that address how human preferences, values, feedback, or norms are represented or aggregated in AI alignment. I excluded work that discusses only technical optimisation without engaging with the question of what is being aligned.

The first question concerns whether the preferentist paradigm, in its current implementations, does what it claims to do. The second concerns whether the difficulties identified can be fixed within machine learning at all, or whether some of them lie outside what any technical method can solve on its own.

Answering them is not the aim of this work. This review does not defend a normative position of its own, nor does it propose an original theory of how AI should be aligned. Its purpose is to examine existing work, explain how the different contributions relate to one another, identify the debates that are currently open, and point out the gaps that remain. The arguments presented here belong to the authors cited; the role of this review is to organise them and to make their connections visible.

1.1 Outline

The review is organised as follows. Section 2 presents the preferentist paradigm: its theoretical assumptions and the two main methods through which it is implemented today, RLHF and DPO. Section 3 examines the criticisms raised against this paradigm, separating conceptual critiques (Section 3.1) from technical and empirical ones (Section 3.2). Section 4 reviews the alternatives proposed in the literature: pluralistic alignment, explicit aggregation through social choice, and approaches based on norms and roles. Section 5 identifies three gaps that the literature leaves open: how to fairly represent heterogeneous populations, how to handle preferences that change over time and depend on context, and how to balance user agency without falling into paternalism or manipulation. Section 6 concludes by returning to the two guiding questions of this review.

2 The Preferentist Approach to AI Alignment

This section introduces how various authors characterise the dominant paradigm in current AI alignment practice, which represents human preferences as utility or reward functions. Section 2.1 presents its theoretical foundations, while Sections 2.2 and 2.3 describe the two main methods through which this paradigm is implemented today: Reinforcement Learning from Human Feedback and Direct Preference Optimisation. This section only describes these assumptions and methods; their critical discussion is left for Section 3.

2.1 Preferences, Utility and Reward

Zhi-Xuan et al. [47] identify four interrelated implicit theses that, taken together, characterise what they call the *preferentist approach* to AI alignment, but these are not necessarily true:

- **Rational choice theory as a descriptive framework.** Human behaviour can be described, at least approximately, as the result of choosing the option that maximises the satisfaction of one’s preferences. These preferences can be represented as a mathematical function (a utility function or reward function) that assigns a number to each available option, where higher numbers correspond to more desirable options.
- **Expected utility (EUT) theory as a normative standard.** It describes how people make decisions and defines what counts as a rational choice. Under this view, a rational agent is one who maximises expected utility. By extension, AI systems are often designed and evaluated according to the same principle.
- **Single-principal alignment as preference matching.** For an AI system to be aligned with a single human user, it should act so as to maximise the satisfaction of that user’s preferences.
- **Multi-principal alignment as preference aggregation.** For an AI system to be aligned with multiple human users, it should aggregate their individual preferences into a single objective and act so as to maximise that aggregated objective.

These four theses are not independent: each one builds on the previous, and together they form a single conceptual foundation. The first two make claims about human beings (how we decide, and how we ought to decide), while the third and fourth translate those claims into design principles for AI systems. As Zhi-Xuan et al. [47, pp. 1814–1816] note, this foundation is rarely defended explicitly in technical work, but it underlies most of the methods currently used to align AI systems with human values.

These assumptions were not invented to design artificial intelligences. They originate in twentieth-century economics and decision theory, where the behaviour of a rational agent is modelled as the maximisation of a mathematical function called a utility function, which assigns a number to each available option to represent how strongly the agent prefers it. The mathematical justification of this framework comes from classical results such as the von Neumann and Morgenstern theorem [35], which shows that consistent preferences under uncertainty can be represented by a utility function [47, Section 2]. Even so, this framework is already an idealised model of human decision-making, since real humans do not always reason or choose in perfectly rational ways. Methods such as Inverse Reinforcement Learning [36] and Reinforcement Learning from Human Feedback [11, 37], discussed in Section 2.2, inherit this assumption when they treat human behaviour or feedback as evidence for an underlying reward function.

As already noted in the introduction, Gabriel [18] observes that the expression “aligning AI with human preferences” can refer to different alignment targets. These targets do not always coincide. Collapsing them into a single notion of preference is already a theoretical choice.

The following subsections show how this conceptual framework has been translated into concrete methods for training language models. Section 2.2 describes Reinforcement Learning from Human Feedback (RLHF), which is currently the most widely used method. Section 2.3 introduces Direct Preference Optimisation (DPO), a recent mathematical simplification that builds on the same paradigm.

2.2 RLHF and IRL as Preference-Based Methods

Currently, two preference-based methods are Inverse Reinforcement Learning (IRL) and Reinforcement Learning from Human Feedback (RLHF) [36, 11]. In IRL, a human or another agent performs the task, and the system tries to infer the reward function that would make this behaviour optimal [36]. RLHF works differently: humans give feedback on the system’s behaviour, often by comparing two options, and this feedback is used to learn a reward function [11, Section 2]. Since comparing two options is much faster than teaching the entire task, RLHF has become especially relevant for tasks where the desired behaviour is difficult to formalise.

The modern version of RLHF has its origins in the work of Christiano et al. [11]. For this review, the point is the basic structure of the method, not all its implementation details. The authors replace a programmed reward with a reward learned from human preferences [11]. The method works as follows:

1. The agent generates behaviours, that is, short sequences of actions in its environment.
2. A human compares pairs of those behaviours and chooses which one they prefer.
3. A neural network learns to predict these preferences and becomes the reward function used to train the agent.

The authors tested the method on Atari games and simulated robotics tasks. Their central result, for the purpose of this review, is that agents could learn complex behaviours while receiving human feedback on less than 1% of their interactions with the environment [11]. Subsequent work extended this approach to language models, first for summarisation [45] and then for general instruction-following [3].

Later, Ouyang et al. [37] applied RLHF to the training of large language models. Their starting point was a simple observation: making a language model larger does not guarantee that it will follow user instructions better. This is because models trained only to predict the next token learn a different objective from the one expected from a useful assistant. They may produce outputs that are false, toxic, or simply unhelpful [37, Section 1].

What changes in this setting is not the basic idea of RLHF, but the training pipeline. Instead of learning from short behaviours in an environment, the model learns from prompts, written demonstrations, and rankings of possible answers. Ouyang et al. [37] first fine-tune GPT-3 on demonstrations written by labellers, then train a reward model on human rankings of model outputs, and finally use this reward model to further fine-tune the model with PPO [37, Section 3.1]. The aim is to make the model more helpful, honest, and harmless, rather than merely better at predicting internet text [37, Section 1].

The result is that a model which is two orders of magnitude smaller but trained with human feedback can outperform a much larger unaligned model in terms of perceived usefulness. However, this depends on the context and specific prompting of the model [37, Section 4.1].

Before concluding this section, it should be noted that the process described above is not a neutral one. Behind each of the three stages, there are a series of decisions that determine the final outcome: who drafts the reference answers and compares the results, what instructions they receive beforehand, what criteria they are asked to prioritise, and how disagreements among the raters are resolved. Lambert,

Gilbert, and Zick [30] highlight that these operational details are rarely made public, even though they are precisely what determines which values end up encoded in the final model [30, Sections 1 and 5]. Section 3 examines the consequences of these decisions in greater detail.

2.3 Direct Preference Optimisation

RLHF, as described in the previous subsection, is a complex process. It requires training a reward model from human preferences and then using reinforcement learning to fine-tune the language model. These challenges led to the search for simpler alternatives. One of the most influential was proposed by Rafailov et al. [39] under the name Direct Preference Optimisation, or DPO. The central idea is to train the language model directly from comparisons of human preferences, in a single step that more closely resembles standard supervised learning. With this approach, no separate reward model is trained. Instead, the language model itself acts as an implicit reward model, thanks to a mathematical equivalence that the authors derive in the paper [39, Section 4]. Subsequent work has proposed related variants such as Identity Preference Optimisation (IPO) [1] and Kahneman-Tversky Optimization (KTO) [15], which share the same underlying assumptions.

In simple terms, DPO changes the training procedure but not the kind of data from which the model learns. RLHF first learns a reward model from human comparisons and then optimises the policy against that reward model. DPO compresses these two stages into one: it uses pairwise comparisons to make the preferred answer more likely and the rejected answer less likely. This is why the method is technically simpler, while still belonging to the same broad family of preference-based methods.

To test the effectiveness of the method, the authors evaluate DPO on three tasks [39, Section 6]. In all three, DPO matches or exceeds the performance of PPO-RLHF despite being substantially simpler to implement [39, Section 6].

Although DPO is technically simpler than RLHF, it does not move away from the theoretical assumptions described in Section 2.1. It still relies on pairwise comparisons between responses and assumes the existence of a latent reward function that explains human preferences [39, Section 4]. When used with preference data from multiple users, it can also inherit the aggregation problems discussed in later sections. The difference from RLHF is therefore mainly operational: DPO reduces the implementation cost but does not transform the conceptual foundation. Gözl, Haghtalab, and Yang [19] confirm this point from a formal perspective, showing that, within their preference learning model with diverse users, DPO and RLHF produce exactly the same distortion [19, App. E.3]. Consequently, the criticisms examined in Section 3 apply equally to both methods.

3 Criticisms of the Preferentist Paradigm

The previous subsections have described the assumptions of the preferentist paradigm and the concrete methods based on it, RLHF and DPO. This section examines the criticism that recent literature has raised against that paradigm. This criticism is divided into two groups. Section 3.1 addresses the conceptual criticisms: those that question whether preferences are the right kind of object to represent human values, and whether the models that sustain preferentism are adequate as descriptions of human behaviour or as normative standards of rationality. Section 3.2 addresses the technical and empirical criticisms: those that show concrete problems in preference-based methods, ranging from the quality of human feedback to the loss of pluralism observed empirically after RLHF training.

3.1 Conceptual Criticism

This subsection examines three conceptual criticisms of the preferentist paradigm. The first is that preferences may be too thin a representation of what people value. The second is that the rational choice and expected utility models behind preferentism are limited both descriptively and normatively. The third is that the expression “aligning with human preferences” remains ambiguous, because it can refer to different alignment targets.

3.1.1 Preferences as a Thin Representation of Values

One conceptual criticism of the preferentist approach concerns the concept of preference itself. In many technical settings, a preference is represented as a comparison between two options: A is preferred to B. This representation is useful because it is simple and easy to collect, but it leaves out *why* A is preferred to B. It does not show whether the preference is based on honesty, usefulness, fairness, clarity, politeness, or some other reason [47, Section 2.3].

The same point appears outside AI alignment. In the broader literature on preferences, Hausman [22] presents preferences as a concept used across economics, psychology, ordinary language, and philosophy, and explicitly relates preferences to choice, value, and welfare [22, pp. ix–xii]. He also notes that preferences are shaped by beliefs, desires, emotions, intentions, reasons, and values, which supports the idea that a preference comparison does not by itself exhaust the content of a human judgement [22, p. xii].

Zhi-Xuan et al. [47, Section 2.3] make a similar point in the context of AI alignment. They claim that preferences are not basic units of human value, but are often constructed from deeper reasons, commitments, and evaluative concepts. A scalar reward model inherits this limitation. It can represent that one response received a higher score than another, but it does not by itself represent the reasons behind that judgement. Related concerns about treating preferences as a single scalar are raised in surveys of RLHF problems [9, Section 3.2].

This does not mean that all technical representations of preferences must be equally thin. For example, CP-nets were introduced to represent conditional *ceteris paribus* preference statements, in other words, preferences that depend on specific conditions or features of the options [8]. This shows that preferences can be represented in richer ways. But it does not solve the deeper problem. Even a more structured preference representation still has to decide which reasons, values, or conditions should count in the first place [47, Section 2.3]. Similar concerns about the limits of one-dimensional preference representations are raised by Conitzer et al. [12, Section 1].

3.1.2 Rational Choice and Bounded Rationality

The first thesis of preferentism holds that human behaviour can be approximated as utility maximisation. In machine learning, this idea is often formalised with the Boltzmann rationality model. This model introduces flexibility by assuming that humans do not always pick the best option. Instead, the probability of choosing an option increases with its utility [47, Section 2].

However, this flexibility is not sufficient. The Boltzmann model still treats human errors as random deviations from optimal behaviour. Psychological evidence gives a different picture: human deviations are often systematic and predictable, not merely random noise [47, Section 2.1]. This is also supported by the tradition of bounded rationality. Simon’s account of satisficing describes how people often look for sufficiently good options rather than optimal ones [41]. Prospect theory shows that decision-making under risk systematically departs from expected utility theory [25]. Framing effects further show that

choices depend on how options are presented [46]. Similar concerns about the irrationality assumptions in preference learning appear in subsequent work on AI alignment [6, 28].

This has direct implications for alignment. Methods such as IRL and RLHF assume that human behaviour or feedback reveals a latent reward function. But if human deviations are systematic, what these methods recover may be a mixture of values, cognitive limitations, framing effects, and constraints on attention or information [47, Section 2.1].

3.1.3 Expected Utility as a Normative Standard

The second thesis of preferentism holds that being rational consists of maximising expected utility, that is, choosing the option with the highest expected value under uncertainty. This connects with the earlier discussion of utility and reward in Section 2.1. If rational agency is understood in terms of expected utility maximisation, then alignment is naturally framed as the problem of learning or specifying the right utility function [47, Section 3].

The classic defence of this idea is based on coherence arguments, especially money-pump arguments. According to this line of reasoning, an agent whose preferences violate certain consistency principles can be exploited through a sequence of exchanges that each seem acceptable locally but together leave the agent worse off [20, Section 1]. This supports the conclusion that a rational agent should have coherent preferences, and that these preferences can be represented by a utility function [47, Section 3].

Yet, this conclusion is weaker than it seems. Recent work on AI and expected utility theory argues that avoiding exploitation does not require every advanced AI system to act as an expected utility maximiser [4, Sections 5–6]. Zhi-Xuan et al. [47, Section 3.1] make the same general point: an agent may avoid some forms of exploitation without satisfying all the axioms required by expected utility theory.

Even if coherence arguments were sound, expected utility theory would still have a limit. It tells an agent how to act given a set of preferences, but it does not tell us whether those preferences are good, ethical, or worth satisfying. For alignment, this matters because a perfectly coherent AI system could still pursue harmful or antisocial goals. In this sense, expected utility theory may help with consistency, but it is not sufficient as a full normative standard for alignment [47, Section 3.3].

3.1.4 Ambiguous Targets of Alignment

The previous criticisms questioned the descriptive and normative models that underpin preference-based systems. This third criticism points to something more fundamental: the very idea of “aligning with human preferences” is ambiguous.

As introduced in Section 2.1, Gabriel [18, Section 3] distinguishes several possible goals of alignment: following explicit instructions, capturing the user’s intentions, mimicking observed behaviour, inferring the preferences a person would have if better informed, promoting their interests, or conforming to their moral values. These targets do not always coincide. The central problem with methods such as RLHF is that a human comparison between two responses is treated as one preference signal, even when it may reflect different kinds of judgement.

The training process can then combine these different judgements into a single reward function. Ouyang et al. [37] explicitly note this limitation in the case of InstructGPT: their method aligns the model to the stated preferences of a specific group of labellers and researchers, rather than to any broader notion of “human values” [37, Section 5.2]. Therefore, the problem goes beyond data quantity or annotation quality. The question of what an AI should align with is a normative question that cannot be resolved

solely from observed preferences [18, Section 3]. Other authors frame this ambiguity in terms of which standards should govern a system in a given context [26, Section 4].

The three criticisms discussed in this subsection converge on the same idea. First, preferences are too limited a representation of what people value. Second, the models underpinning preferentism fail both as descriptions of human behaviour and in their role as normative standards. Third, the very notion of “aligning with preferences” is ambiguous and lumps together distinct phenomena that current methods fail to distinguish. The fourth thesis of preferentism, which reduces multi-principal alignment to the aggregation of individual preferences, is examined in Section 3.2.

3.2 Technical and Empirical Criticism

This subsection turns from the conceptual problems to the technical and empirical ones that arise when the preferentist paradigm is implemented in methods such as RLHF and DPO. The aim is to show how the philosophical objections appear in practice. The discussion is organised around three problems: the difficulty of obtaining reliable human feedback, the limitations of reward models and preference aggregation, and empirical evidence that current alignment methods can reduce pluralism.

3.2.1 Problems of Human Feedback

The first major problem is obtaining good human feedback. Casper et al. [9, Section 3.1] distinguish several challenges in the feedback stage of RLHF, including biased evaluator selection, difficulty of oversight, data quality problems, and limitations in the type of feedback collected. The model does not learn human preferences in general. It learns from the particular people selected as evaluators, under particular instructions, in a particular work context.

One issue is that annotators may not represent the diversity of users and affected groups. Data pipelines can introduce bias if evaluators are selected because they agree with researchers, or if disagreement among annotators is treated as noise rather than as evidence of genuine variation [9, Section 3.1.1]. Annotator disagreement has been studied directly as a source of systematic variation rather than noise [16, 5]. Another issue is that even well-intentioned evaluators can be affected by fatigue, time pressure, limited attention, or lack of expertise. In difficult tasks, they may overlook serious errors or be unable to evaluate a response reliably [9, Section 3.1.2].

There is also no perfect feedback format. Pairwise comparisons are easy to collect, but they do not show how strong a preference is. Scalar scores can express intensity, but different annotators may use the same numerical scale in different ways, which makes calibration difficult. Natural language feedback can provide richer information, but it is harder to convert into a reward model [9, Section 3.1.4]. In short, human feedback is not a neutral window into human values. It is already shaped by the method used to collect it.

The case of InstructGPT provides a concrete example of this problem. Ouyang et al. [37, Sections 3.3 and 3.4] trained their models using demonstrations and comparisons produced by around 40 contractors, mostly working in English-language tasks. The authors explicitly note that the resulting model is aligned with the stated preferences of a specific group of people, mainly their labellers and researchers, rather than with any broad notion of “human values” [37, Section 5.2].

The final behaviour of the model therefore reflects several layers of influence: the prompts used for training, the labellers who produced demonstrations and rankings, the instructions given to them, and the researchers who designed the study and answered edge-case questions. Ouyang et al. [37, Sections 5.2 and 5.3] also acknowledge remaining limitations: InstructGPT still makes mistakes, can produce harmful

or biased outputs, and may follow user instructions even when doing so is undesirable. The point is not that this makes the method useless. It shows that RLHF depends on a socially situated feedback process, not on direct access to human values.

3.2.2 Problems of the Reward Model and Preference Aggregation

Even if human feedback were reliable, the reward model would raise separate problems. RLHF compresses human feedback into a reward model, but human values and judgements are not always easy to represent as a single scalar function. Casper et al. [9, Section 3.2.1] connect this to problem misspecification, criticism of the reward hypothesis, and the difficulty of modelling human irrationality. The authors also report agreement rates of approximately 63% to 77% in previous RLHF studies, both among annotators and between annotators and researchers [9, Section 3.2.1]. This suggests that disagreement is not only noise, but may reflect genuine variation in values, priorities, and interpretations. If feedback from many people is condensed into one reward model without modelling these differences, the problem becomes ill-specified [9, Section 3.2.1]. A further difficulty is that reward models are usually evaluated indirectly: a policy is trained against the reward model, and then the policy is evaluated. This makes it hard to separate the quality of the reward model from the quality of the optimisation process [9, Section 3.2.3]. Related failure modes of reward learning have been formalised as reward hacking and reward misspecification [43, 38].

A related problem is hidden context. Siththaranjan, Laidlaw, and Hadfield-Menell [42, Section 1] define hidden context as information that affects a preference judgement but is not represented in the data used to train the reward model. This can include annotator identity, cognitive limitations, or different annotation objectives. Their Pell Grant example illustrates the point: if most users of a college-admissions assistant are high-income students, the system may learn that information about low-income financial aid is less preferred, even though this information is essential for low-income students [42, Section 1]. Social choice theory studies how to aggregate individual preferences or judgements into collective decisions, and provides a long tradition of analysing voting rules and their properties [48]. A social choice rule is a method for taking individual rankings and producing a collective choice or ranking. The Borda count is one such rule: it assigns each alternative a score based on its position in each individual’s ranking, and selects the alternative with the highest total score [12, Section 2.5]. The Borda count has known weaknesses: it ignores the intensity of preferences (a strongly preferred alternative counts the same as a slightly preferred one) and it is vulnerable to manipulation through strategic ordering of alternatives [48]. Siththaranjan, Laidlaw, and Hadfield-Menell [42, Section 3] show that, in the specific case of RLHF, when hidden context is present standard preference learning implicitly aggregates comparisons using a Borda-type rule. In that pairwise setting, the consequence is that diverse judgements are collapsed into one aggregate score, which can favour majority patterns and obscure minority needs.

The technical proposal of Siththaranjan, Laidlaw, and Hadfield-Menell [42] is Distributional Preference Learning, or DPL. Instead of learning one score for each response, DPL learns a distribution of possible values, making it possible to represent that a response may be good in one context, bad in another, and uncertain in a third [42, Section 4]. The authors also introduce a metric to estimate how much variation is explained by observed features and how much is left to hidden context [42, Section 4]. They apply this idea to Helpful and Harmless RLHF (HH-RLHF) data, where some comparisons concern helpfulness and others concern harmlessness. In this setting, the annotation target itself can become hidden context if the model is not told which objective is being prioritised [42, Section 5]. The authors also connect this to jailbreaks, that is, prompts designed to bypass a model’s safety restrictions. Their results suggest that DPL can detect some hidden context, and that more risk-averse optimisation can reduce vulnerability

to this kind of failure [42, Section 5].

The same aggregation problem is studied from a different angle by Gölz, Haghtalab, and Yang [19]. Their question is minimal: even if a method does not represent diversity perfectly, does it at least serve users well on average? They introduce the concept of distortion to measure this loss. Distortion compares the best average utility that could be achieved if the training process had access to users’ actual utilities with the average utility achieved by the learned policy [19, Section 1]. The paper also introduces the idea of a *mythical user*. When heterogeneous comparisons are fitted into one Bradley-Terry model, the reward model does not represent a real user. It represents a fictional user who best explains the aggregate data [19, Section 1]. In simpler terms, the model is optimising for someone who is nobody. In the version of their model without the AI-specific KL constraint, RLHF coincides with a Borda-type rule; in their formal analysis, DPO is equivalent to RLHF and therefore suffers from the same distortion [19, App. E.3]. This matters because the problem is not unique to one implementation of RLHF. It affects a broader family of methods based on aggregate preference optimisation.

These results connect RLHF to social choice theory. Conitzer et al. [12, Sections 1 and 2.5] argue that social choice should guide AI alignment because alignment with diverse feedback raises questions about whose feedback is collected, what criteria are used, and how conflicting judgements are aggregated. A related line of work responds by changing the optimisation objective itself. For example, Chakraborty et al. [10, Section 1] propose MaxMin-RLHF, which aims to improve outcomes for worse-off preference groups rather than only optimising an aggregate reward. The point is simple: preference aggregation is not a neutral technical detail. Once feedback comes from heterogeneous users, the aggregation rule helps determine whose preferences count, whose are averaged away, and which groups may be disadvantaged.

3.2.3 Empirical Evidence of Reduced Pluralism

The previous paragraphs identified several technical problems: human feedback is socially situated, reward models compress heterogeneous judgements into a single signal, and preference aggregation can disadvantage minority preferences. A final concern is empirical. If RLHF is supposed to align models with human preferences, then post-training should not make models less representative of the diversity of those preferences.

Sorensen et al. [44] provide preliminary evidence that this can happen. In their analysis, post-RLHF models are less close to distributions of human responses than their base models. In other words, after alignment training, the models become less representative of the variation found in human answers [44, Section 5.2]. This result is important because it connects the formal concerns discussed above with measurable changes in model behaviour.

A similar concern appears in empirical work on whose opinions language models reflect. Santurkar et al. [40, Section 4] show that models do not equally represent all demographic groups, and that reinforcement learning from human feedback can shift model outputs toward particular groups’ opinions rather than simply making them more representative overall. Durmus et al. [14] extend this concern to global opinions, arguing that measuring alignment requires asking how well models represent subjective opinions across different countries and populations. These studies support the same general point: alignment affects safety, helpfulness, and the distribution of perspectives represented by the model. Related empirical studies have measured political and cultural skew specifically [21].

In light of the above, RLHF has both formal and empirical limits. The problem goes beyond difficult feedback or misspecified reward models. Preference-based optimisation can also compress disagreement and reduce pluralism. The process that is supposed to align the model with human preferences can

therefore move it further away from the distribution of those preferences [44, Section 5.2].

4 Alternative Approaches

After the critical assessment in the previous section, other authors turn to possible solutions to the problems raised. This section reviews three overlapping responses to these problems: pluralistic alignment, explicit aggregation through social choice, and approaches that shift the target of alignment toward norms and roles.

4.1 Pluralistic Alignment

As noted in Section 3.2, Sorensen et al. [44] argue that current alignment techniques can reduce distributional pluralism. Their paper is also constructive. It proposes a positive alternative: pluralistic alignment. The idea is that human users have diverse values and perspectives, so a well-aligned system should serve more than an average user. It should be able to represent, adapt to, or preserve that diversity. The authors formalise three ways in which a model can be pluralistic [44, Section 3]:

- **Overton pluralism.** The model presents the spectrum of reasonable responses to a question, rather than choosing a single response as if it were neutral. For example, if asked about a controversial topic such as content moderation on social media, an Overton-pluralist model would respond by describing the different reasonable positions (public safety, freedom of expression, combating disinformation).
- **Steerable pluralism.** The model can be directed to reflect certain attributes, perspectives, or communities, within an acceptable range of behaviour. In other words, the user or the developer can ask the model to respond from a particular point of view, and the model does so faithfully.
- **Distributional pluralism.** The distribution of the model’s responses matches the distribution of responses from a reference population. For example, if in a survey 45% of people say one thing, 32% say another, and 9% say a third, a distributionally pluralist model would produce responses with approximately those frequencies.

In addition to the three types of pluralism in models, the paper proposes three corresponding types of benchmarks, that is, ways to evaluate a model’s pluralism [44, Section 4]:

- **Multi-objective benchmarks.** A model must fulfil several objectives to produce what would be considered a good response. Rather than evaluating the model with a single score that encompasses all these objectives, it is given an individual score for each one. This makes the trade-offs visible, in other words, what is gained and what is lost in each individual objective.
- **Trade-off steerable benchmarks.** These evaluate whether the model has the ability to change its behaviour when asked to prioritise one objective over another. For example, if asked to prioritise more safety and less freedom, does the model respond by coherently adjusting?
- **Jury-pluralistic benchmarks.** The model is evaluated by different judges with different perspectives. Rather than averaging their scores, the individual score from each judge is preserved to retain the diversity of their responses.

In Section 2, Sorensen et al. [44] give several reasons why pluralism should be an explicit goal of alignment. First, no single default model can fit all users equally well, so some form of customisation may be needed. This connects to the single-principal assumption discussed in Section 2.1: in practice, a one-size-fits-all model often behaves as if there were one default user, even when the model is used by many different

people. Second, treating human variation as noise means losing valuable information. Third, foundation models are used in many different contexts, so a single aggregate evaluation gives an incomplete picture. Finally, in democratic societies, representing a diversity of perspectives is itself a value that AI systems should respect.

However, each form of pluralism has limits. Sorensen et al. [44, Sections 3.1–3.3] discuss these limits separately for Overton, steerable, and distributional pluralism. These sections suggest that no form works equally well in all contexts. The right form depends on what the model is used for, who uses it, and what kind of disagreement is at stake. Empirical work also shows that current models do not equally represent demographic groups, which makes the choice of reference population non-trivial [40, Section 4].

Personalisation has been proposed as a more concrete path toward this kind of pluralism. Kirk et al. [27] describe traditional alignment as a one-size-fits-all model, where all users receive the same model with the same default values. This connects again to the single-principal assumption: the model behaves as if there were one generic user, or one default set of preferences, even though real users differ. This can exclude minority groups or impose a dominant culture. Personalisation appears as one possible response to this problem. The authors define personalised alignment as the conditional adaptation of a model’s responses to a specific user. In simple terms, the model responds to the text in front of it and adapts to the user’s preferences, needs, or context [27, Section 1]. The paper distinguishes two main paths:

- **Explicit personalisation** occurs when the user directly expresses what they want, for example, “respond briefly” or “I prefer simple explanations”.
- **Implicit personalisation** occurs when the system infers preferences from conversation patterns, documents, history, behaviour, or personal data.

Building on this distinction, the authors highlight both benefits and risks. At the individual level, personalisation can increase efficiency, autonomy, and empathy. A personalised model may require fewer prompts, adapt better to the user’s level of knowledge, and make the interaction feel closer. But the risks mirror the benefits. A model that adapts too well can create dependency, reinforce existing biases, or produce algorithmic profiling. At the social level, personalisation can support inclusion and accessibility, but it can also produce unequal access, polarisation, echo chambers, and personalised political or commercial manipulation [27, Section 3]. Concrete personalisation methods have been proposed by Jang et al. [24] and Li, Lipton, and Leqi [32].

In Section 4, Kirk et al. [27] move from benefits and risks to limits. They note that it is not enough to let each user personalise the model as they wish, because some customisations can cause social harm. They therefore propose hierarchical limits: legal restrictions for gravely harmful content, organisational limits set by companies or communities, and a zone of permitted personalisation for lower-risk aspects such as style, complexity, tone, memory, or conversational preferences [27, Section 4].

The same problem appears in pluralistic alignment more generally. Distributional pluralism requires deciding which population should serve as the reference. Overton pluralism requires deciding which responses count as reasonable. Steerable pluralism requires deciding which dimensions of behaviour users should be able to adjust. The point is not that alignment could ever avoid normative choices. It cannot. The point is that pluralistic alignment makes those choices visible. Someone still has to decide who counts, what counts as reasonable, and where the limits of adaptation should be.

Personalisation can therefore be read as a concrete form of steerable pluralism. It does not solve the problem by itself. It only changes the question. Instead of asking which values the model should have by default, it asks which forms of adaptation should be allowed. Together, Sorensen et al. [44] and Kirk

et al. [27] show the same thing: managing human diversity in alignment is technical and normative at once.

4.2 Social Choice and Explicit Aggregation

The previous subsection argued that pluralistic alignment still requires normative choices. Social choice theory offers one way to make those choices more explicit. In simple terms, social choice theory studies how individual preferences or judgements can be combined into a collective decision. This is directly relevant to AI alignment. If different people give different feedback, the system still has to decide how that feedback is combined.

In this context, Conitzer et al. [12] argue that aligning AI systems with diverse human feedback can be understood as a social choice problem. Collecting more feedback is not enough. The harder question is how to combine judgements that may be different or even contradictory. Methods such as RLHF already do this, but they usually do it implicitly and without a clear justification [12, Section 1].

From the perspective of social choice, the RLHF evaluation system raises several key questions: which humans give feedback, what criteria they use, how their judgements are combined, whether those expressed judgements are truly their preferences, and what happens when different groups disagree. As explained in Section 2.2, when data comes from diverse sources, training a single reward model necessarily implies a form of preference aggregation that often remains opaque and poorly justified. This problem does not disappear with approaches such as Constitutional AI. Constitutional AI is a method in which a model is trained to follow a set of written principles, or a “constitution”, rather than relying only on direct human labels for every output [2, Section 1]. This changes where the aggregation problem appears. Instead of asking only whose preferences are learned from feedback, we must also ask who writes the constitution, what principles it includes, how conflicts between principles are resolved, and how social diversity is represented [12, Section 1].

For all these reasons, the authors propose turning to social choice theory. This field studies problems such as preference aggregation, minority representation, voting rules, strategic manipulation, and judgement aggregation across different criteria. These are not separate from alignment. They are already present when human feedback is used to train a single model. Social choice theory does not remove disagreement, but it provides tools to describe it and to make explicit which aggregation rule is being used [12, Section 1].

This can apply to principles or criteria as well as to people. For example, a response could be evaluated according to safety, usefulness, and fairness. The problem then becomes how to combine these evaluations into a final decision [12, Section 7.4].

This proposal connects directly with several results already discussed in Section 3, and brings them under a common framework. In Section 3.2, the discussion of Siththaranjan, Laidlaw, and Hadfield-Menell [42] showed that, when hidden context is present, preferences are aggregated following a rule equivalent to the Borda count. In the discussion of Gözl, Haghtalab, and Yang [19], a similar conclusion was reached but from a different perspective: when training a single reward model with comparisons from heterogeneous users, RLHF also behaves like a Borda rule, and ends up optimising for a mythical user that does not correspond to any real person. Despite starting from different problems, both studies make the same point: a technical optimisation process is also a particular way of aggregating opinions, with its own rules and limitations.

This is where the framework from Conitzer et al. [12] clarifies the problem. The authors argue that, if RLHF already resembles a social choice rule, this choice of rule should be made explicit. The Borda rule is not the only possible one; it is just one option among several, and it has consequences, such as

not taking into account how strong a preference is or being vulnerable to manipulation. For this reason, they propose not replacing RLHF, but making visible that a decision of this kind is already being made and discussing it openly [12].

A recent example makes this point clearer. Maura-Rivero et al. [33] propose replacing standard RLHF aggregation with maximal lotteries, a probabilistic voting rule from social choice theory. The idea is that the system should not simply optimise a scalar reward learned from aggregate comparisons. It should use an aggregation rule with clearer social choice properties. The authors also connect this to Nash Learning from Human Feedback (NLHF), because NLHF-style methods approximate maximal lottery outcomes [33]. This is useful for the argument here because it shows what the proposal by Conitzer et al. [12] can mean in practice: social choice can guide the design of new alignment rules in concrete ways.

The common lesson is simple: aggregation cannot be treated as a neutral technical step. RLHF, Constitutional AI, Borda-type aggregation, and maximal lotteries all combine different inputs in different ways. Each choice has consequences. It affects whose views are represented, how disagreement is handled, whether minority positions are preserved, and whether strategic behaviour is possible. If these choices are hidden inside an optimisation process, the system can look neutral while still making normative decisions. The point of a social choice perspective is to make those decisions visible.

4.3 Norms, Roles and Principles Instead of Preferences

Some of the alternatives discussed above still begin from preferences or judgements. Another response shifts the target itself. Instead of asking only how preferences should be aggregated, it asks which standards should govern a system in a given role. Section 3.1 already showed how Zhi-Xuan et al. [47] criticise the conceptual basis of the preferentist paradigm. In the positive part of their work, the authors propose an alternative: to stop aligning systems with preference aggregation and align them with normative standards [47, Section 4]. In short, each system has a different role, and each role has an associated ideal of how it should behave. A conversational assistant, for example, should not align with the sum of its users' preferences. It should align with what a good assistant is supposed to be: useful, honest, respectful, and not deceptive. The goal stops being what people want and becomes what the role requires.

The authors also propose seeing AI systems as tools, not as agents that pursue their own goals [47, Section 3.2]. A tool is evaluated by whether it does its job well in the context in which it is used, not by whether it satisfies aggregated preferences. This view also makes room for normative reasoning frameworks. Instead of encoding values as a number to be maximised, the system can work with norms and constraints that belong to its role [47, Section 3.3]. Overall, the proposal is not to find the right utility, but to identify the standards a system should respect given the place it occupies. This connects with a broader sociotechnical literature on AI alignment [31].

This idea of orienting alignment toward norms instead of preferences connects with the positive proposal of Gabriel [18], which was already mentioned in Section 2.1 from its critical side. The starting point of Gabriel [18] is not that we should find the true moral theory and put it into machines. The aim is to find fair principles for alignment that can receive reflective support in plural societies [18, Section 4]. This is not a pure break with social choice. The principles still need public support. The difference is that what is being selected is not a model response, but the principles that should guide the system. The clearest practical example of this approach is Collective Constitutional AI, discussed below in this subsection [23, Sections 1 and 3].

To answer this question, Gabriel [18] explores three paths taken from political theory:

- The first is overlapping consensus, which looks for principles that different moral traditions can accept, even if for different reasons; human rights are the clearest example.
- The second is the veil of ignorance, an idea adapted from Rawls. The question here is which principles people would choose if they did not know who they were or what position they would occupy in relation to the technology.
- The third is democratic processes. Here principles gain legitimacy through debate, voting, and citizen participation, not because they are morally correct in the abstract, but because of the support they receive.

The three paths share the same goal: to find a fair way of deciding the principles of alignment that does not consist in imposing the view of some on others. For this reason, Gabriel’s approach is close to social choice at the design stage [18, Section 4]. Rather than aggregating preferences for each output, the approach asks how principles for the system should be selected in the first place.

A clearer break with preference aggregation appears in Kasirzadeh and Gabriel [26]. The authors focus on conversational agents and argue that different conversations have different standards. A good scientific answer, a good contribution to democratic debate, and a good creative exchange cannot be judged by the same single criterion.

They call these standards discursive ideals. In science, the relevant ideals include accuracy and coherence. In democratic debate, they include civility, sincerity, and mutual respect. In creative exchange, the standards are more open and flexible [26, Section 4]. The point is simple: each domain has its own norms.

This connects to the hidden context problem discussed in Section 3.2. If the quality of a response depends on the domain, then a single reward model can mix judgements that are not really comparable. A better answer may require asking which standards apply in the situation, not just aggregating preferences more carefully.

The approaches discussed in this subsection share a common idea. They do not treat preferences as the final answer to alignment. Zhi-Xuan et al. [47] frame the issue in terms of roles and normative standards. Gabriel [18] frames it in terms of principles that can be publicly justified. Kasirzadeh and Gabriel [26] frame it in terms of discursive ideals proper to each domain. In all three cases, “the question becomes less what do people prefer?” and more “what standards should govern this system here?”

However, this does not remove the need for collective decision-making. Someone still has to decide which roles, principles, and norms matter. Social choice can therefore appear at two levels. It can appear inside the system, when feedback is aggregated in RLHF, DPO, or related methods. It can also appear before the system is trained, when people decide which standards the system should follow. This is why the norms-and-roles approach is not completely separate from social choice. It moves the question to a different level.

A concrete example is Collective Constitutional AI. Huang et al. [23] propose a process where public input is used to create a constitution for a language model. The process includes selecting a target population, collecting principles from participants, transforming those principles into a constitution, and then training and evaluating the model [23, Sections 1 and 3]. This shows that collective decision-making can happen before the model is trained. The issue is not limited to aggregating feedback inside RLHF. It also concerns who helps define the standards that guide the system in the first place. Foundational discussions of how AI systems should be embedded in social and institutional contexts appear in Bommasani et al. [7].

These approaches therefore show why alternatives to preferentialism do not escape normative decisions.

They make those decisions more explicit. The question is no longer hidden inside a reward model. It becomes a question about roles, principles, institutions, and participation. Those political questions are precisely what the next section examines.

5 Open Questions and Future Directions

The previous sections have shown the same pattern several times. Current methods face technical problems and force normative choices: whose preferences count, how disagreement is handled, and when user control should be limited. The alternative approaches reviewed above make these choices more visible, but they do not remove them.

This section therefore does not repeat the details of the earlier discussion. It identifies three open questions that remain after reading the literature: how to represent heterogeneous populations (Section 5.1), how to model preferences that change over time and across contexts (Section 5.2), and how to give users agency without turning personalisation into manipulation or paternalism (Section 5.3).

5.1 Gaps in the Representation of Heterogeneity

The first open question concerns heterogeneity. Sections 3.2 and 4 showed that current methods often compress diverse feedback into one signal. The problem appears in different forms. Siththaranjan, Laidlaw, and Hadfield-Menell [42] describe it as hidden context. Gözl, Haghtalab, and Yang [19] describe it as optimisation for a mythical user. For Sorensen et al. [44], the result is a loss of pluralism. The open question is no longer whether diversity matters. It is how it should count.

This question should not be reduced to pairwise comparisons alone. Social choice theory offers richer feedback formats, including rankings and approval ballots [48]. Computational preference modelling also includes structured models such as CP-nets [8, p. 136]. This matters for alignment because different feedback formats preserve different kinds of information. A pairwise comparison may say that A is preferred to B, but it does not show how strongly, for whom, or under which conditions.

One future direction is therefore to compare alignment methods by average utility and by their effects on different groups. MaxMin-RLHF is one example, because it changes the objective toward worse-off groups [10]. Maximal lotteries offer another example from social choice theory [33]. Distributional Preference Learning points in the same direction by making hidden context visible [42]. Future work could ask which aggregation rules protect minorities, which rules are vulnerable to manipulation, and which rules make sense for different domains. The hard part is partly technical. Someone still has to decide what fairness means in the first place. A direct call to apply distortion analysis to RLHF appears in Dai and Fleisig [13].

5.2 Gaps in the Temporal and Contextual Dimension

The second open question concerns time and context. Preferences are not fixed objects. They depend on reasons, situations, roles, and domains. This has already appeared in the discussion of hidden context [42]. It also appears in the discussion of discursive ideals [26]. Personalisation raises the same issue from the user side [27]. The open question is how an alignment method should respond when preferences change.

This problem also appears outside AI alignment. Social choice theory has already studied repeated and dynamic decision-making. Iterative voting studies situations where voters can revise their choices over time [34]. Perpetual voting studies long-term collective decision-making, where the history of previous

decisions matters for fairness [29]. This literature does not offer direct solutions for AI alignment. But it suggests a useful direction: alignment should ask what people prefer at one moment, how preferences evolve, whose preferences have been repeatedly ignored, and when the history of interaction should affect the next decision.

For AI alignment, this raises several research questions. Should a model remember past user preferences, or should it treat every interaction as new? Should a system adapt when a user changes their mind, or should it protect the user from unstable or harmful short-term preferences? Should feedback collected in one context be used in another? Current work does not yet answer these questions. It mainly shows why they matter.

5.3 Gaps Concerning User Agency

The third open question concerns user agency. Personalisation promises more control for the user, but also creates new risks. Kirk et al. [27, Sections 3 and 4] show this clearly: the same mechanisms that make a model more useful can also create dependency, reinforce bias, enable profiling, or support manipulation. The open question is where the limits should be. Lazar and Nelson [31] frame this question in sociotechnical rather than purely technical terms.

This question cannot be answered by saying that the user should always decide. As Gabriel [18, Section 3] argues, aligning with instructions is not the same as aligning with intentions, interests, or values. A user may ask for something that harms them or others. But the opposite answer is also dangerous. If the model or its designers decide what is best for the user, agency moves away from the person using the system.

Future research should therefore focus on the boundary between control and protection. One useful direction is to separate low-risk personalisation from high-risk personalisation. Style, tone, memory, and level of explanation may be easier to leave to the user. Political persuasion, medical advice, self-harm, financial decisions, or harmful requests need stronger limits. Value-sensitive design is relevant here because it treats values as part of the design process, not as something added after the system is built [17]. Collective Constitutional AI points in a similar direction, by showing how public input can be used before training to shape the principles of a language model [23].

The remaining question is practical and normative at the same time. Who should define the limits of user agency? The user, the developer, the state, affected communities, or some combination of all of them? The literature reviewed in this thesis does not give a final answer. It shows why the question cannot be avoided.

5.4 Assessment

The three gaps point to the same conclusion. Future work can improve preference optimisation. It also has to study the procedures through which preferences, principles, and limits are selected. What I take from the literature is that the main gap is procedural. The question is not only which preferences to learn, but how the process of learning, contesting, updating, and limiting alignment should be governed.

In my view, the most promising underdeveloped direction is to treat alignment as a repeated and contextual process, rather than as a one-off task of learning a static preference signal. This direction connects the three gaps discussed above. It asks whose preferences have been ignored over time, how users should be allowed to revise or contest the system’s assumptions, and who should decide the limits of adaptation.

Existing work gives pieces of this picture, but not yet a unified framework. Iterative voting shows how

collective decisions can involve revision over time [34]. Perpetual voting shows how decision history can matter for fairness toward minorities [29]. Value-sensitive design shows why values should be considered as part of the design process, not added after the system is built [17]. Collective Constitutional AI shows one possible way of involving public input before model training [23].

This suggests a research direction that is both technical and political: alignment methods should not only optimise preferences, but also make room for revision, contestation, and institutional decision-making. The problem is how to train the model, and how to decide what kind of alignment process we want.

6 Conclusion

This literature review started from two questions. The first was whether current alignment methods capture or respect real human preferences. The second was whether better methods can solve the remaining problems, or whether there are deeper limits in the attempt to represent diverse, changing, and conflicting preferences through machine learning. The previous sections have unfolded the answer in three stages: Section 2 described the dominant paradigm, Section 3 examined its limitations, and Sections 4 and 5 discussed the alternatives proposed in the literature and the gaps that remain. The two questions can now be answered.

The answer to the first question is limited. Current methods do capture preferences, but only in a narrow and mediated sense. They capture feedback produced by particular people, under particular instructions, in particular contexts. They do not capture human values directly. Gabriel [18, Section 3] shows why this matters: alignment can target instructions, intentions, preferences, interests, or values, and these are not the same thing. Zhi-Xuan et al. [47, Section 2.3] make the related point that preferences are too thin to represent the full content of what people value. In this sense, preference-based alignment does not remove the problem of values. It translates it into a reward model.

The answer to the second question is more complicated. Better methods can improve parts of the problem. Distributional Preference Learning can make hidden context more visible [42]. Pluralistic alignment can preserve more information about disagreement and diversity [44]. Social choice can make aggregation rules explicit [12]. These are real improvements. But they do not remove the central difficulty. Someone still has to decide whose views count, how conflict should be handled, what standards should guide the system, and where the limits of user control should be drawn.

This is why the alternatives discussed in this review should not be treated as completely separate paths. Pluralistic alignment, social choice, personalisation, constitutional approaches, and norm-based approaches overlap. Social choice can be used inside the system, for example when feedback is aggregated [12]. It can also be used before the system is trained, when people decide which principles or standards should guide it [23]. Norms and roles do not replace collective decision-making. They move it to another level.

The main finding of this review is therefore not that preference-based methods are useless. They are useful, and they have become central to current language model alignment. The problem is that they are often asked to do more than they can do. They are technical methods for learning from feedback. They are not, by themselves, a theory of what AI should align with.

The open questions follow from this. Future research needs to ask how heterogeneous populations can be represented without simply averaging away minorities. Gözl, Haghtalab, and Yang [19, Section 6] frame this as the question of whether alignment should measure average utility and equity. Future research also needs to ask how much agency users should have over personalisation. Kirk et al. [27, Section 4] show

that the limits of personalisation are normative as well as technical. These questions require machine learning, but also social choice theory, political philosophy, ethics, and human-computer interaction.

The central conclusion is simple. Alignment cannot be solved only by collecting more preferences or optimising them more efficiently. The question of what an AI system should align with is also a question about power, representation, and justification. Better algorithms can help. They cannot decide this question on their own.

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